

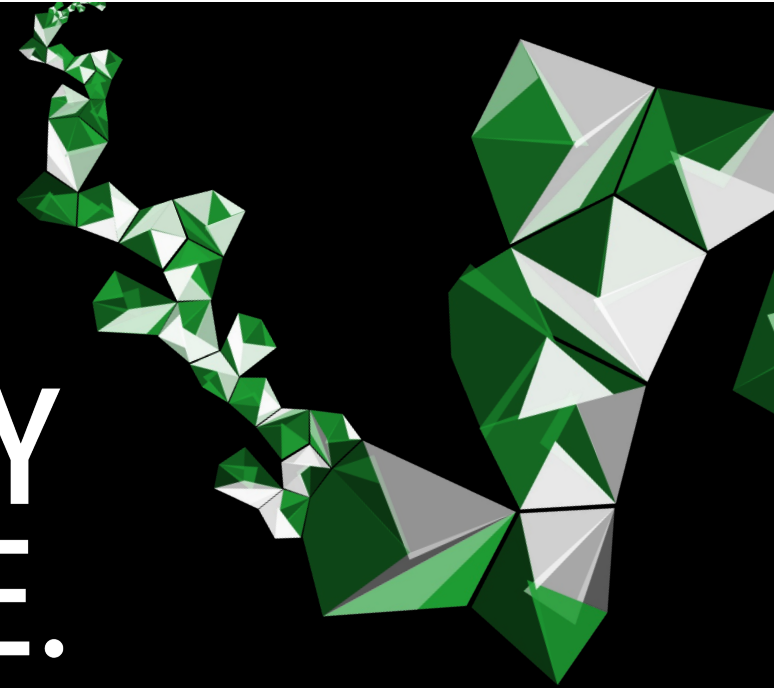
FACULTY OF ELECTRICAL ENGINEERING, MATHEMATICS AND COMPUTER SCIENCE (EEMCS)

UNIVERSITY OF TWENTE.

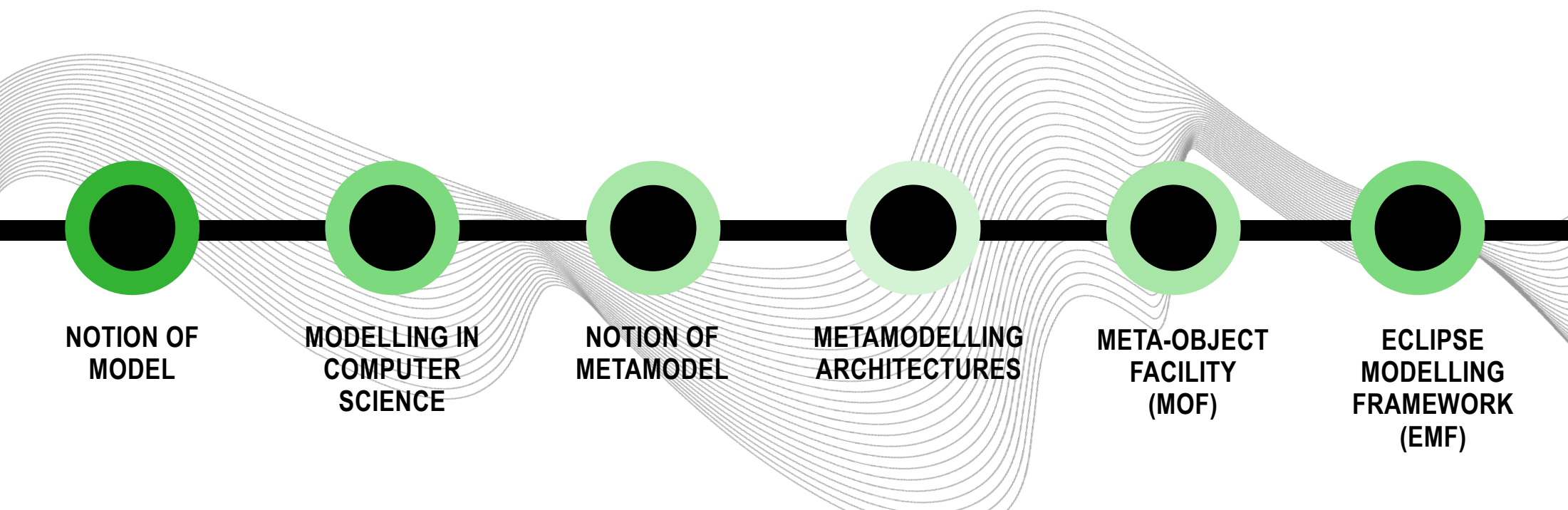
LECTURE 1: MODELS AND METAMODELS

192135450 MODEL-DRIVEN ENGINEERING

31 AUGUST 2026



IN THIS PRESENTATION:



**NOTION OF
MODEL**

**MODELLING IN
COMPUTER
SCIENCE**

**NOTION OF
METAMODEL**

**METAMODELLING
ARCHITECTURES**

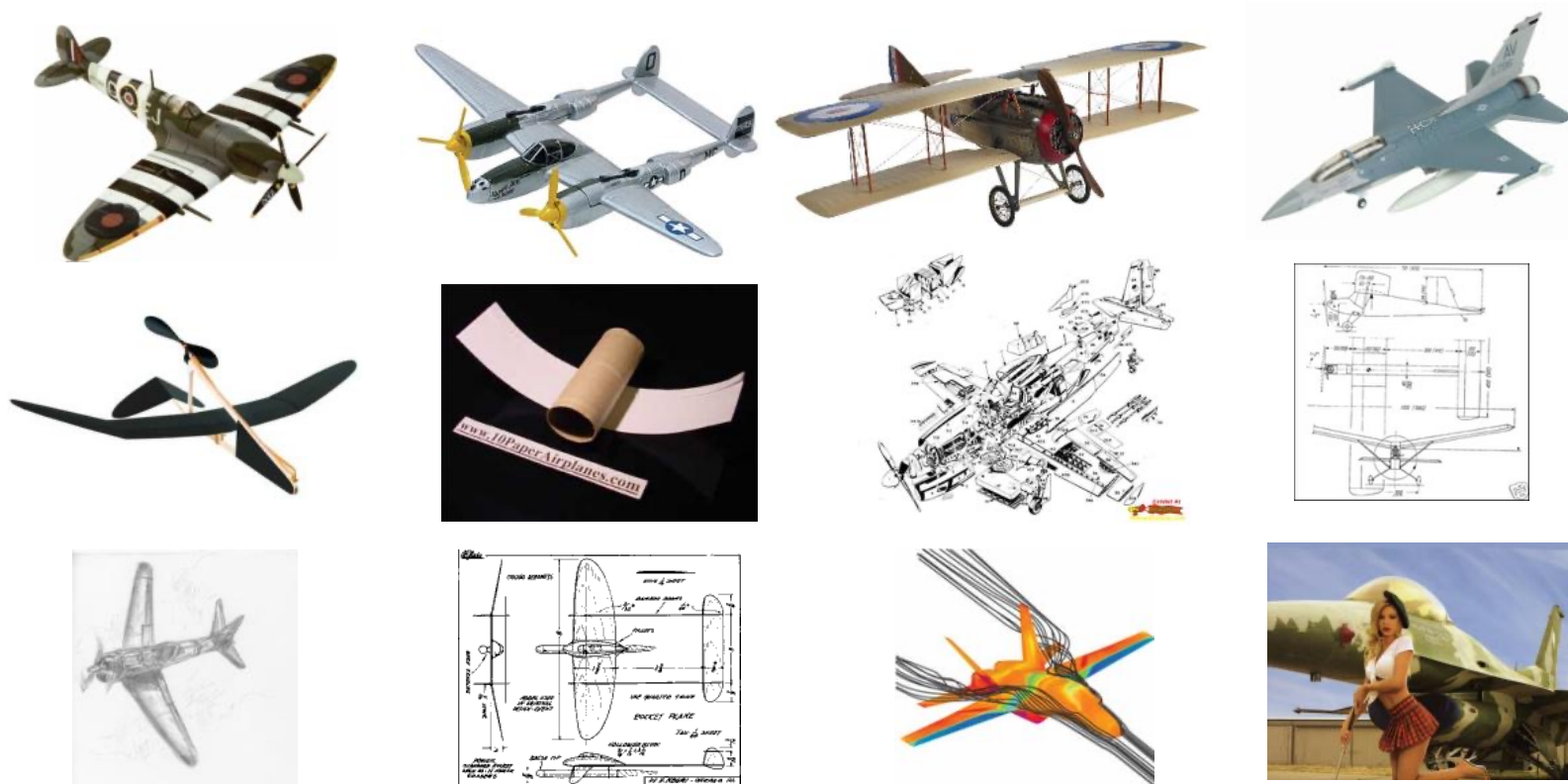
**META-OBJECT
FACILITY
(MOF)**

**ECLIPSE
MODELLING
FRAMEWORK
(EMF)**

WHAT IS A 'MODEL'?



'AIRPLANE MODEL' IMAGES IN GOOGLE



NOTION OF MODEL

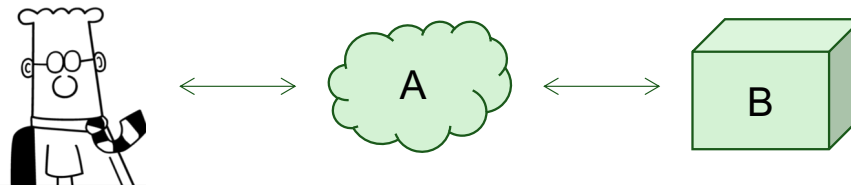
WHAT IS A MODEL?

- In Latin: 'modulus' means 'small measure' (first used in 1575)
- Merriam-Webster online gives 14 meanings to the noun 'model' (see , amongst others
 - “Miniature representation of something; also a pattern of something to be made”
 - “Example for imitation and emulation”
- Different science and engineering disciplines give different definitions of model adapted to their specific context and purposes

MODELS IN PHILOSOPHY OF SCIENCE

DEFINITIONS

“Any subject using a system A that is neither directly nor indirectly interacting with a system B to obtain information about the system B, is using A as a model for B” *Leo Apostel, 1960*



“The object of study in modelling is the real world, where the observer-researcher attempts to detect patterns of recurring relationships which can be represented systematically. To model is to represent this pattern or relationships in a manner which can lead itself to a formal study”

John van Gigch

MODELS IN COMPUTER SCIENCE

DEFINITIONS



“A model is a purposely abstracted, clear, precise and unambiguous conception”

“A model denotation is a precise and unambiguous representation of a model, in some appropriate formal or semi-formal language” *FRISCO Report**

“A model is a representation of a concept. The representation is purposeful: the model purpose is used to abstract from the reality the irrelevant details” *Starfield, Smith and Bleloch, 1990*

* A Framework of Information Systems Concepts,
<http://www.mathematik.uni-marburg.de/~hesse/papers/fri-full.pdf>

MODELS IN COMPUTER SCIENCE

MORE DEFINITIONS



“A model of a system is a description or specification of that system and its environment for some certain purpose. A model is often presented as a combination of drawings and text. The text may be in a modelling language or in a natural language” *MDA Guide v1.0.1*

“A model is information selectively representing some aspect of a system based on a specific set of concerns” *MDA Guide rev. 2.0*

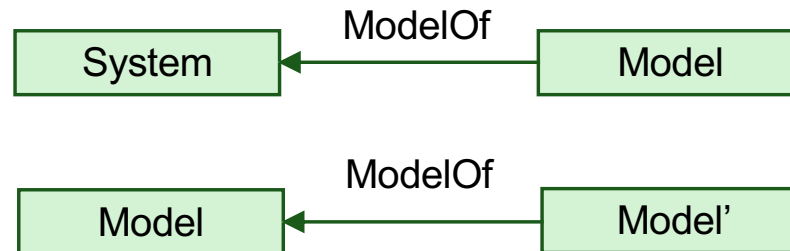
“A model is a simplification of a system built with an intended goal in mind. The model should be able to answer questions in place of the actual system” *Bézivin and Gerbe, 2001*

(GENERAL) CHARACTERISTICS OF MODELS

- 
- Representation of something in the real world (some system)
 - Simplification (abstraction)
 - Conception or concrete representation
→ concrete representation is necessary!
 - Purpose: often descriptive, prescriptive or predictive
 - Desired qualities: precise, unambiguous, allows analysis

SYSTEM AND MODEL

- A **model** requires a **part of the real world** that is modelled
 - System being modelled
 - **ModelOf** relation
- Model can be seen as a **role**
 - A model may **also be the subject of modelling** → 'Model of a model'



MODELS AS ABSTRACTIONS

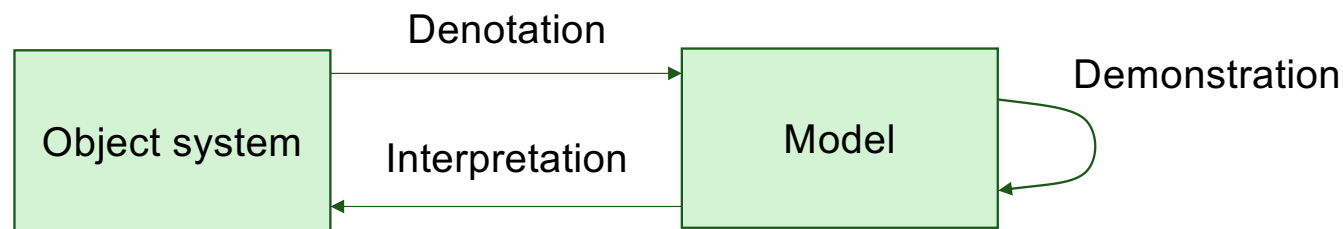
- Abstraction is a powerful cognitive tool for mastering complexity!
- In models, some of the characteristics of the reality are ignored (abstracted from)
 - The purpose of the model guides the abstraction process in which models are produced!
 - Models are abstractions!



Which characteristics are represented and abstracted from in this model?

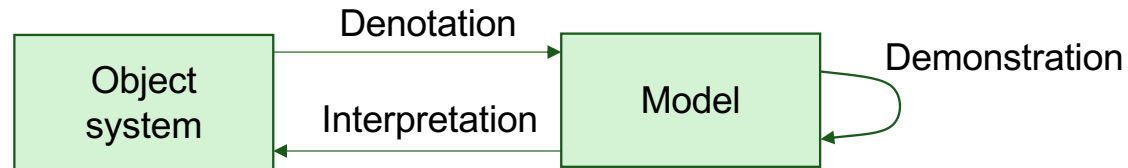
NATURE OF THE MODEL OF RELATION

- **Denotation**: some of the properties of the system are represented or denoted in the model
- **Demonstration**: knowledge is obtained from the model in the terms of the model elements
- **Interpretation**: the obtained knowledge is translated in terms of the system



DDI account (Hughes 1999)

WORKING DEFINITION OF MODEL



- A model
 - represents a part of the reality called the object system
 - is expressed in a modelling language
 - provides knowledge for a certain purpose
- This knowledge can be interpreted in terms of the object system

PURPOSE OF MODELS

[BRAMBILLA, CABOT, WIMMER 2012]

- **Models as sketches**: used for communication, **often partial** (incomplete) views of the object system
- **Models as blueprints**: used to provide **detailed and complete** specification as **prescription** of what should be built
- **Models as programs**: used to **develop the system**, as opposed to code

<http://www.martinfowler.com/bliki/UmlMode.html> discusses how UML can be used in these three modes (and is very critical about MDA 🤔)

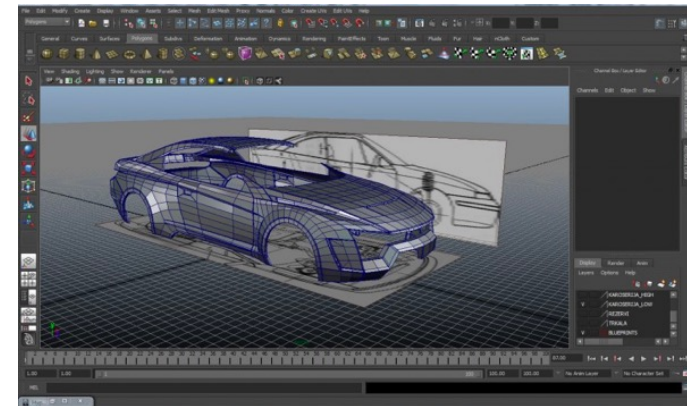
- MDE stresses ‘Models as programs’, without disallowing the others

SOFTWARE SYSTEMS AS MODELS

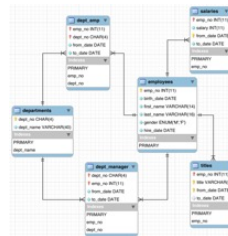
- A running software system can be considered as a model of a system

Examples

- Information system of a university
 - Model of a car in a CAD program
 - Climate simulator
 - Airplane aerodynamic flow simulator
- However, running software may interact with and change the real world
→ Be careful when considering these systems as models!

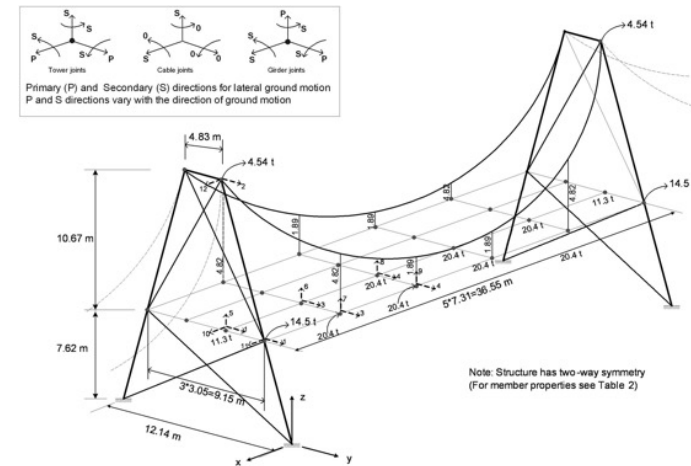


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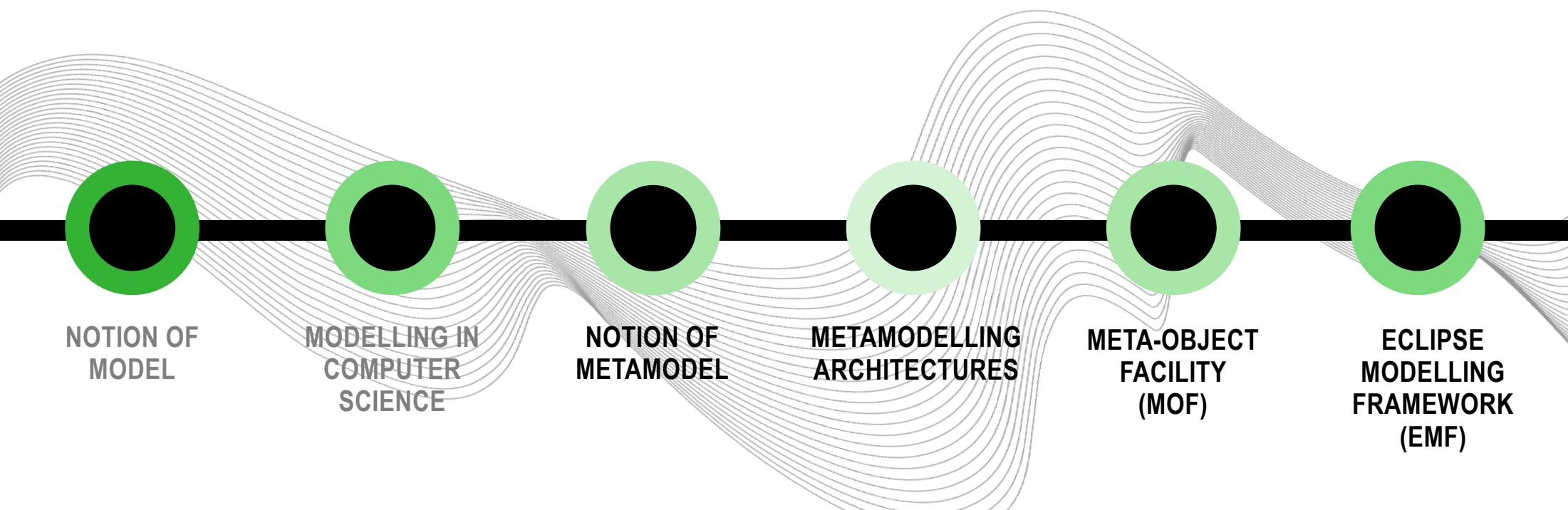


MODELLING IN SOFTWARE ENGINEERING

- Modelling is well-established in many engineering disciplines!!!
- Potential benefits for Software Engineering
 - Productivity increase
 - Reasoning about the system before building it
 - Proving properties of the system at model level
- Ultimately MDE should lead to software with a better quality



IN THIS PRESENTATION:



NOTION OF
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MODELLING IN
COMPUTER
SCIENCE

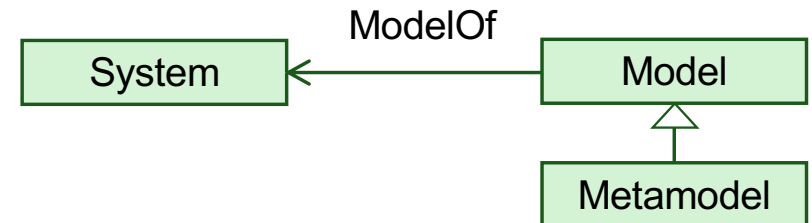
NOTION OF
METAMODEL

METAMODELLING
ARCHITECTURES

META-OBJECT
FACILITY
(MOF)

ECLIPSE
MODELLING
FRAMEWORK
(EMF)

METAMODEL



- A **metamodel** is a **model**, as the name suggests
- What is the **system** modelled by a metamodel?
- What is the **nature of the ModelOf relation**?
 - Which **aspect** of the model is being modelled in its metamodel?
- Why do we sometimes say that a **model** is an **instance of a metamodel**?



METAMODEL AND METAMODELLING

MORE DEFINITIONS

Metamodel as a model of a modelling language

“A meta-model is a model of models expressed in a given modelling language” Ed Seidewitz

“A meta-model is a model of the conceptual foundation of a language, consisting of a set of basic concepts, and a set of rules determining the set of possible models denotable in that language” FRISCO Report

DEFINITION OF METAMODEL

- 
- In MDE, the view that a **metamodel** represents a modelling **language** is widely accepted
 - We assume that a metamodel is a **model of a modelling language**

However

- What is a **modelling language**?
- Which **language components** are **modelled** in the **language metamodel**?

DEFINITION OF METAMODEL

Language is a **set of sentences** (models)

- A metamodel models the **valid members of the set**
- A metamodel **constrains the valid models** expressible in a given modelling language

Analogy with natural languages (English, Dutch, etc.)

- Words allow sentences to be formed
- Language grammar determines which sentences are allowed (the **valid members of the set**)

DEFINITION OF METAMODEL

Language has concrete syntax, abstract syntax and semantics

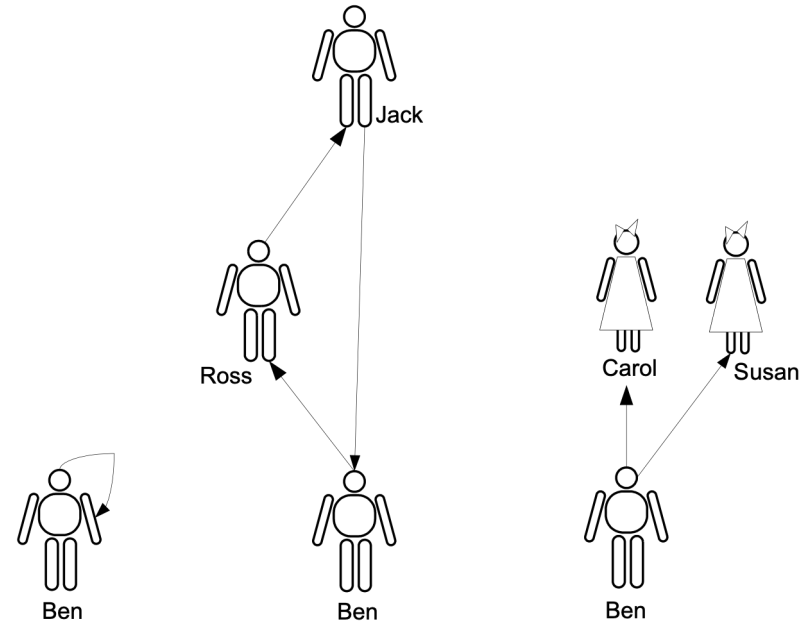
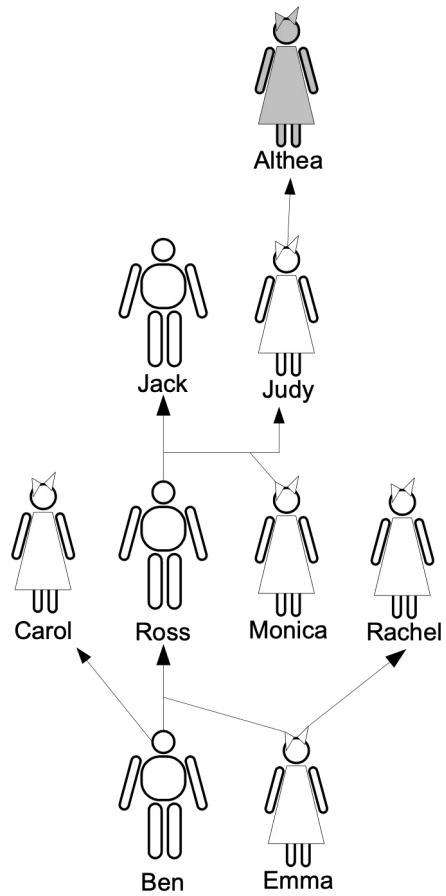
- A metamodel should focus on the concepts that can be expressed with the language and their relationships
- Very roughly corresponds to the language 'abstract syntax' → syntax elements without considering how they are represented

Simplification: metamodel $\approx$ language abstract syntax

EXAMPLE: A GRAPHICAL LANGUAGE FOR GENEALOGY [GUIZZARDI]

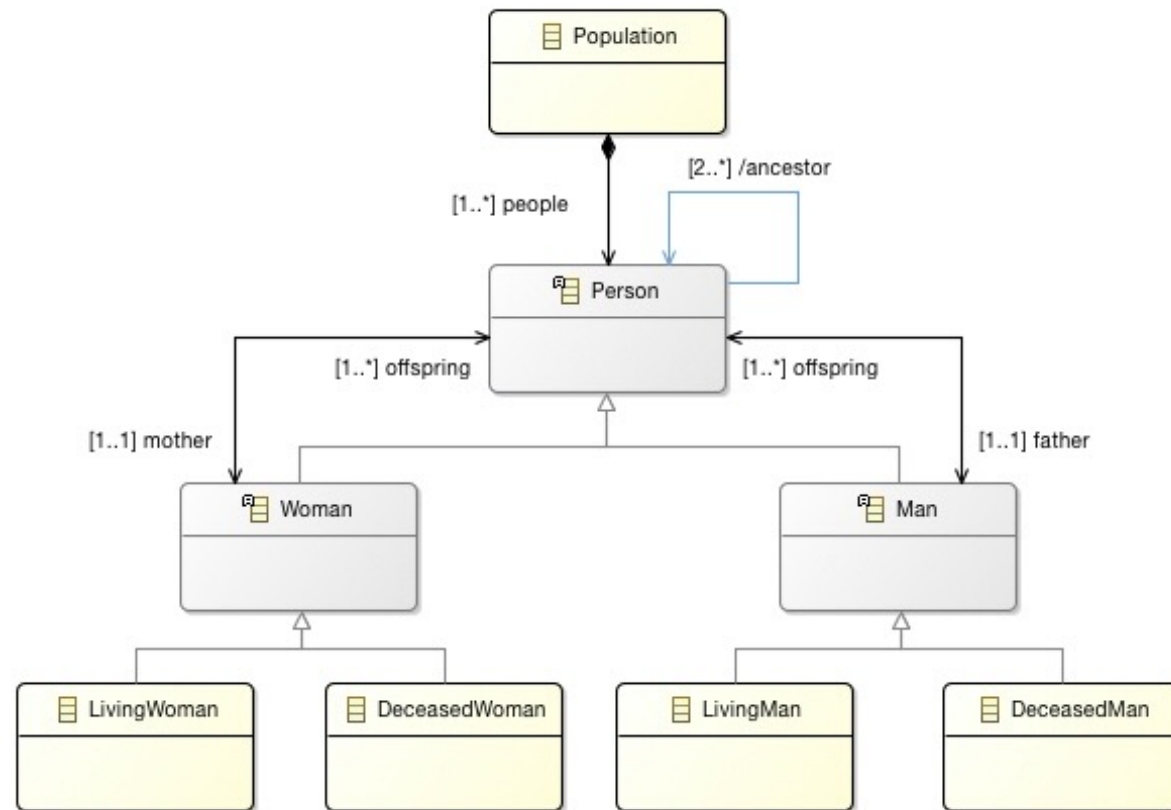
Language $\mathcal{L}_1$									composition of <i>is-arrow-path-connected</i> with the <i>above</i> relation in the plane, e.g.
Ontology	Living Man	Deceased Man	Living Woman	Deceased Woman	ParentOf	Father	Mother	Offspring	AncestorOf

MODEL EXAMPLES



Invalid models

GENEALOGY METAMODEL



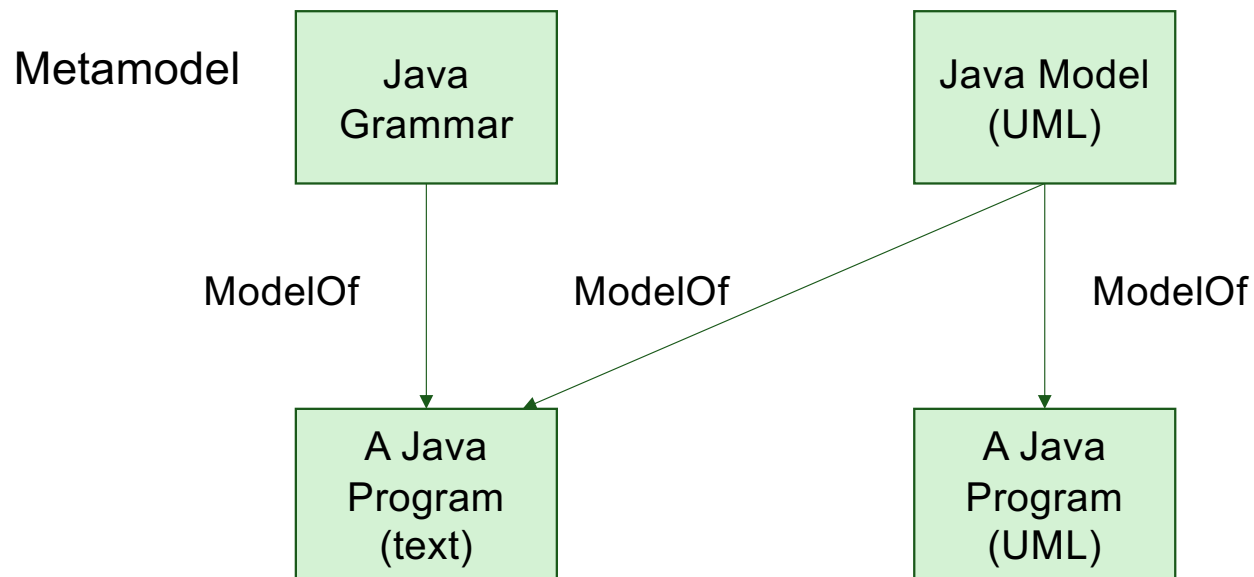
METAMODEL AS A 'MODEL OF A MODEL'

Alternative line of reasoning

- A metamodel can be defined as a **model of a model**
- Different characteristics of a model can in principle be chosen to be modelled in a metamodel
- In MDE, **metamodelling** considers the 'types of the model elements' and consequently the **abstract syntax of the modelling language** as the characteristic represented in the metamodel!

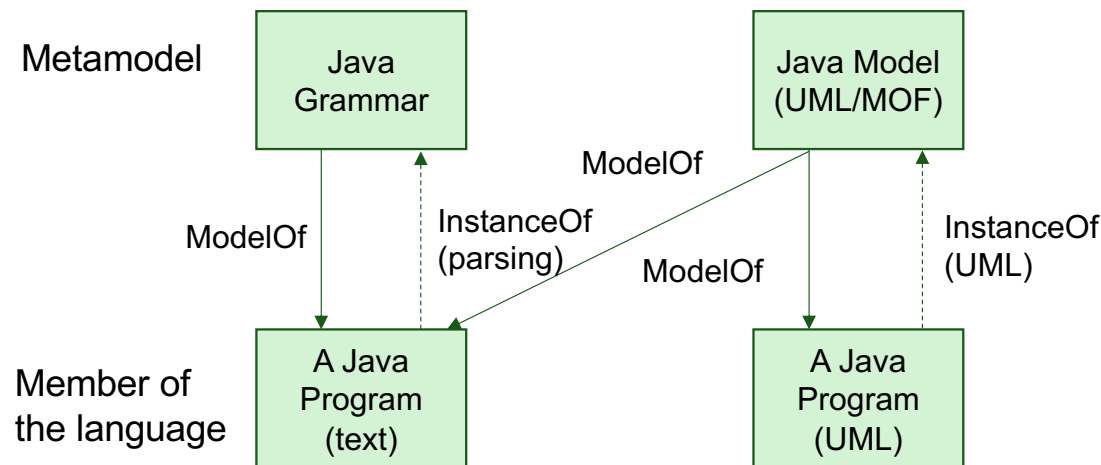
MODELS, METAMODELS, INSTANCE OF RELATION

- A language may have its **metamodel** expressed in different **modelling languages**



MODELS, METAMODELS, INSTANCEOF RELATION

- We often say that a **model is an instance of a metamodel**
- There are different (language-specific) instanceof relations
- InstanceOf allows to **interpret a metamodel 'formally' (systematically)**



METALEVELS

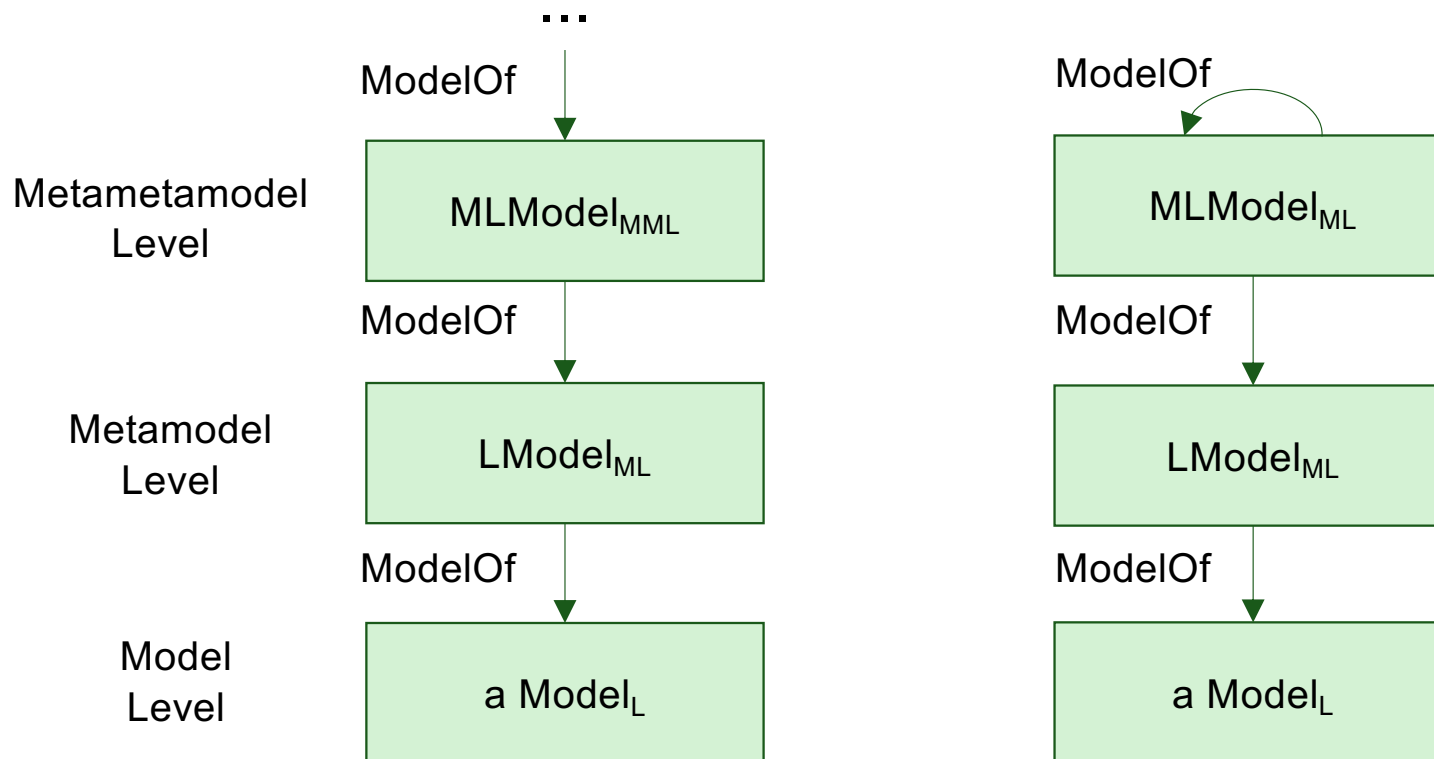
- 
- Model is expressed in a language $\mathcal{L}$
 - Metamodel of $\mathcal{L}$ is expressed in $\mathcal{ML}$
 - Metamodel of $\mathcal{ML}$ is expressed in $\mathcal{MML}$
 - Metamodel of $\mathcal{MML}$ is expressed in $\mathcal{MMMML}$
 - Etc.

How often should or can we repeat it?

METAMODELLING ARCHITECTURES

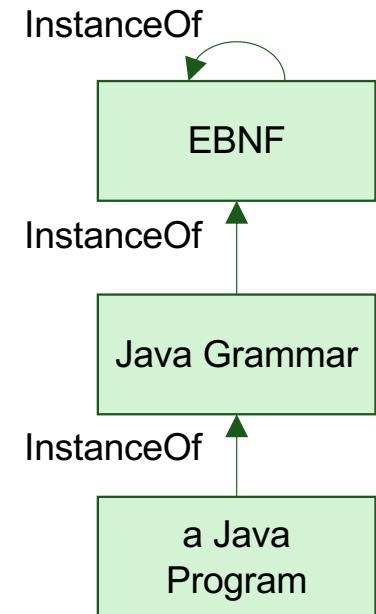
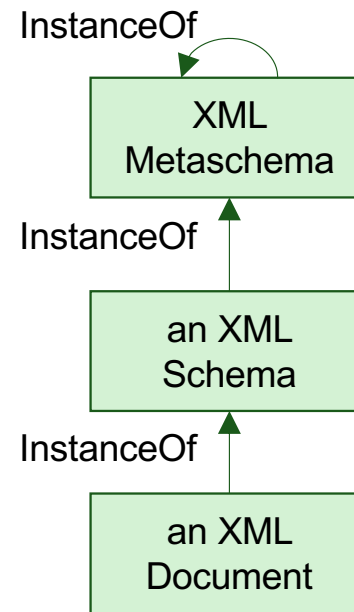
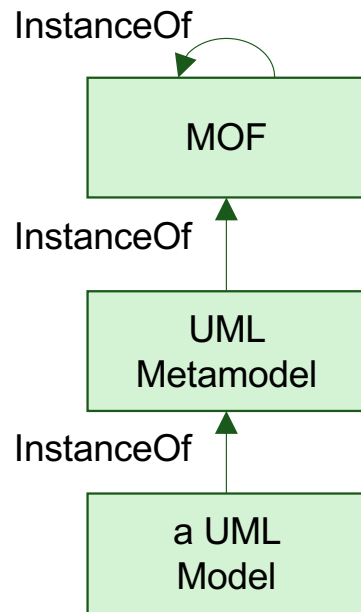
- 
- Two alternative ways to finish the recursive (possibly infinite) tower
 1. Assume that **some language** is just given (e.g., XML DTD)
 2. Model the 'top-language' **in itself**
 - **Option 2** is the most popular in metamodeling architectures

METAMODELLING ARCHITECTURES



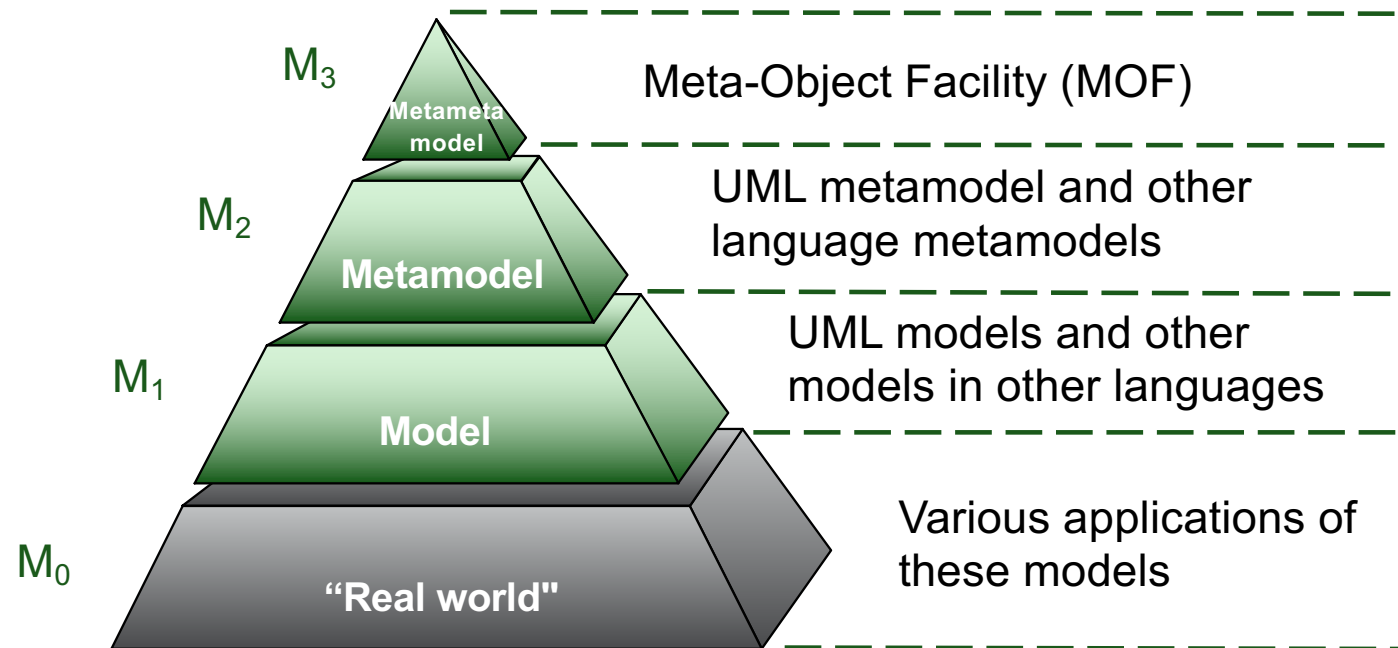
METAMODELLING ARCHITECTURES

EXAMPLES



OMG METALEVELS

META-OBJECT FACILITY (MOF)



OMG METALEVELS: EXAMPLE

M3 (MOF)

Association

Class

<<instanceOf>>

<<instanceOf>>

<<instanceOf>>

<<instanceOf>>

<<instanceOf>>

M2 (UML)

Attribute

0..n

Class

classifier

InstanceSpecification

<<instanceOf>>

<<instanceOf>>

<<instanceOf>>

M1 (User model)

Person

+age : Integer

<<snapshot>>

: Person

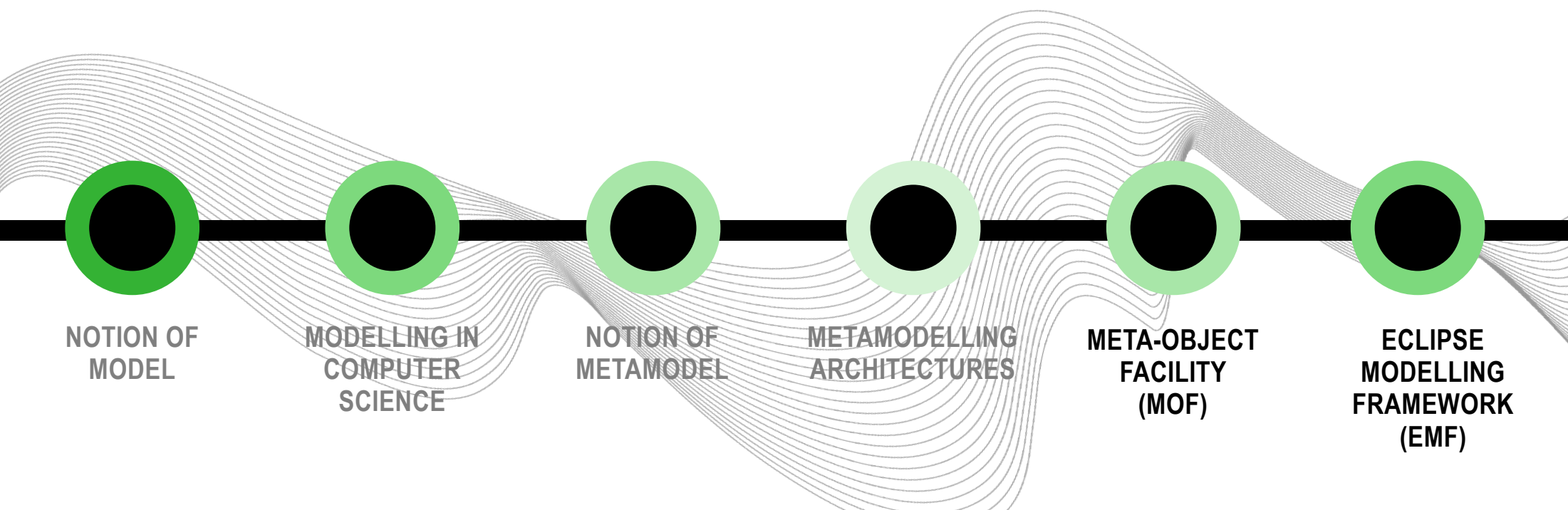
age = 28

<<instanceOf>>

M0 (Runtime instances)

aPerson

IN THIS PRESENTATION:



NOTION OF
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META-OBJECT
FACILITY
(MOF)

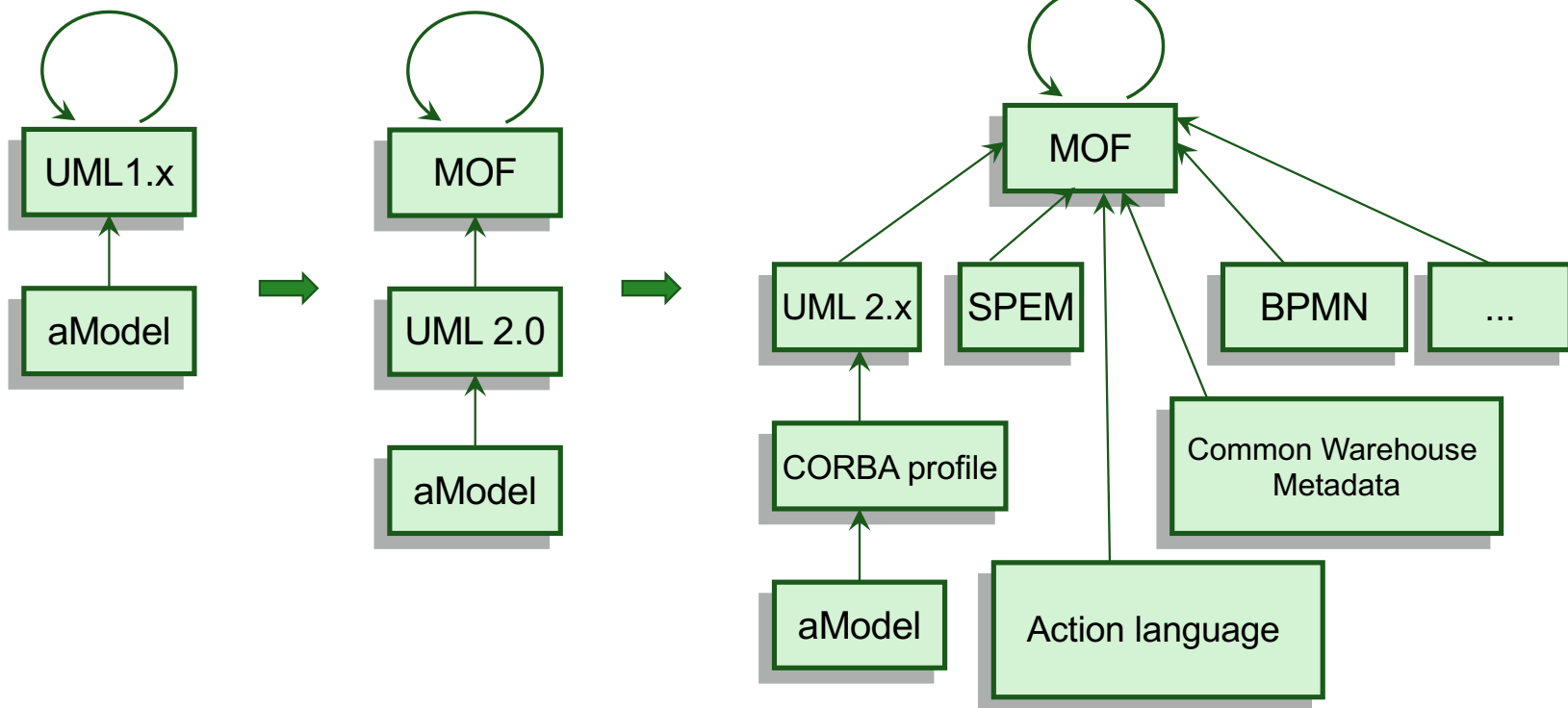
ECLIPSE
MODELLING
FRAMEWORK
(EMF)

META-OBJECT FACILITY (MOF)

- 
- Metadata management framework
 - Language to be used for defining languages
 - OMG-standard metamodeling language
 - UML metamodel is defined in MOF
 - MOF 2.0 shares a common core with UML 2.x
 - Simpler rules for modelling metadata
 - Easier to map from/to MOF
 - Broader tool support for metamodeling
 - alignment allows UML 2.x tool to be used for 'drawing' a metamodel

EVOLUTION OF OMG LANGUAGES

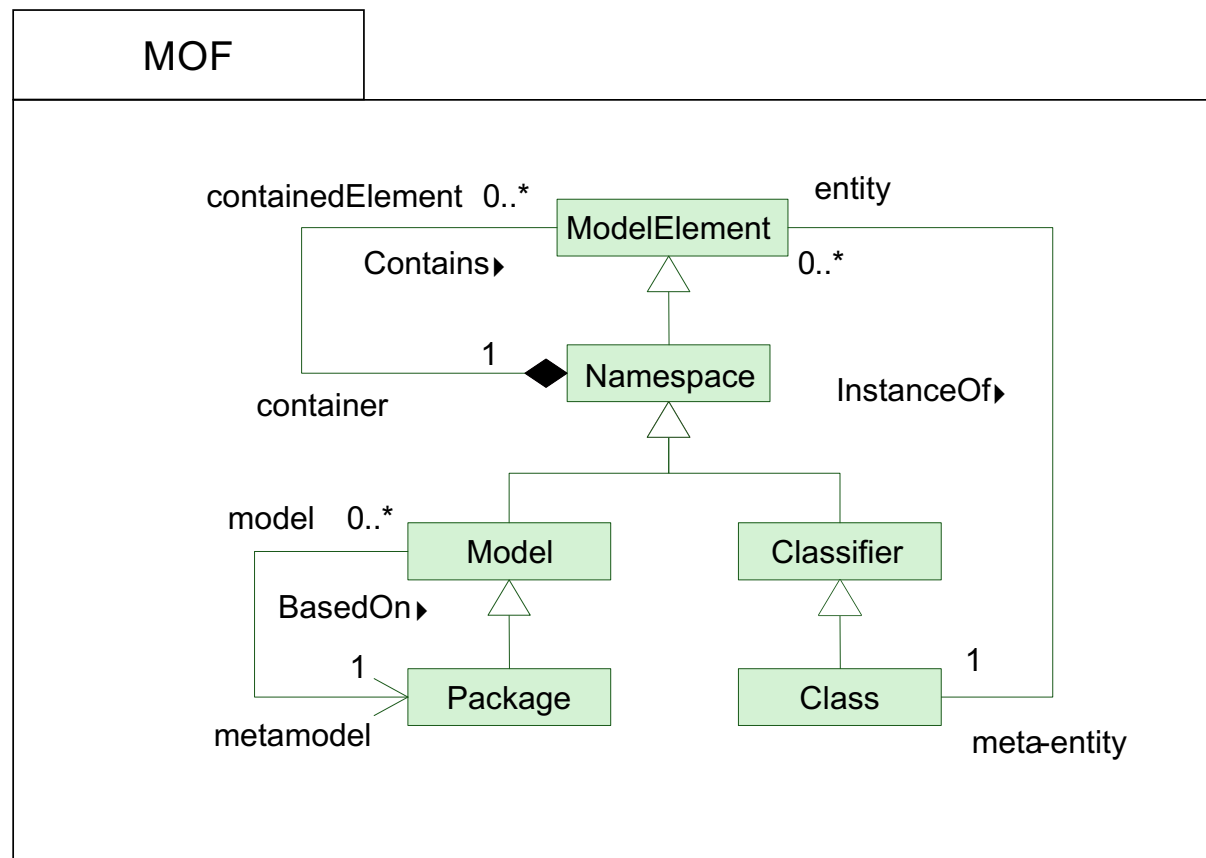
MOF HISTORY



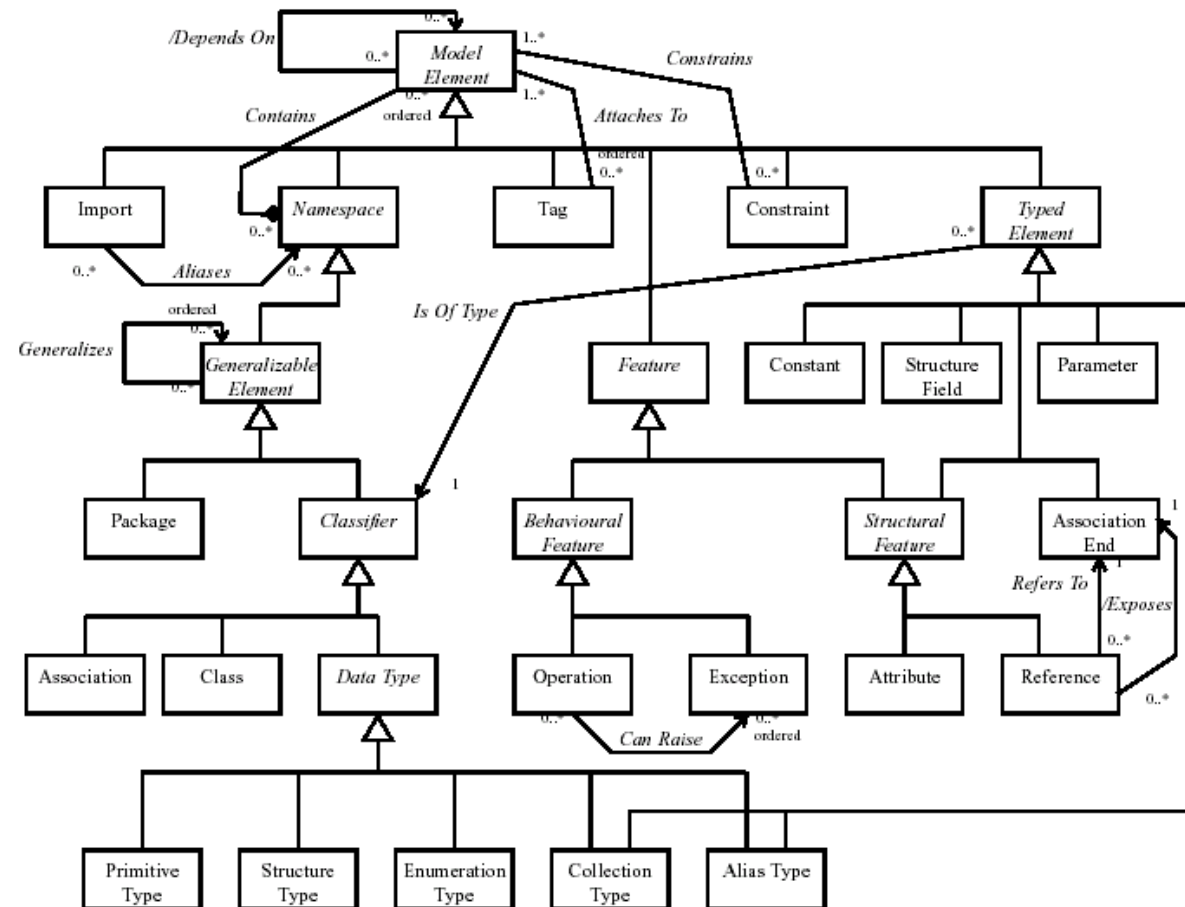
MOF VERSIONS

- 
- MOF 1.x was the **most widely supported** by tools
 - MOF 2.x (2.5.1)
 - **Current standard**, has been **substantially influenced by UML 2.0**
 - Critical for the support of transformations, e.g., QVT and Model-to-text

MOF 1.X MAIN CLASS DIAGRAM



MOF 1.X COMPLETE MODEL



MOF 1.X KEY ABSTRACT CLASSES

NOT DIRECTLY INSTANTIATED, FOR STRUCTURING ONLY

- ModelElement is the **common base Class** of all metaclasses
 - Every ModelElement has a name
- Namespace is the **base Class** for metaclasses that are **containers**
- GeneralizableElement is the **base Class** for metaclasses that **allow generalization** (similar to inheritance in OOP)
- TypedElement is the **base Class** for metaclasses that **have a type** (e.g., Attribute, Parameter and Constant)
- Classifier is the **base Class** for metaclasses that **define types** (e.g., Class and DataType)

MOF 1.X MAIN CONCRETE (META)CLASSES

- 
- Class
 - Association
 - Exception (for defining abnormal behaviours)
 - Attribute
 - Constant
 - Constraint

MOF 1.X KEY ASSOCIATIONS

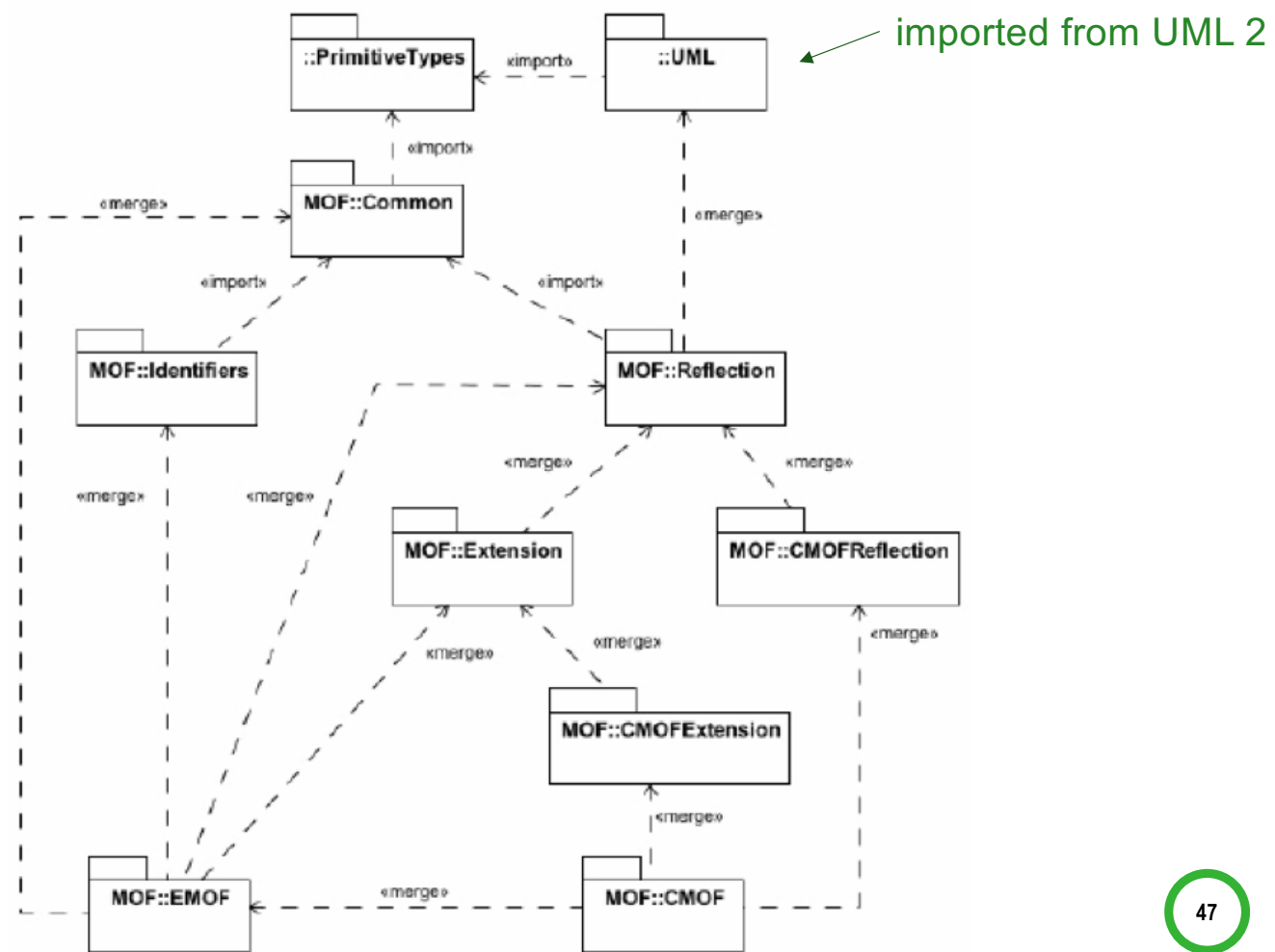
- 
- Contains: relates a **ModelElement** to its **Namespace**
 - Generalizes: relates a **GeneralizableElement** to its **ancestors** (superclass and subclass)
 - IsOfType: relates a **TypedElement** to the **Classifier** of its type
 - DependsOn: relates a **ModelElement** to **another ModelElement** to represent dependence

MOF 2.X STRUCTURE

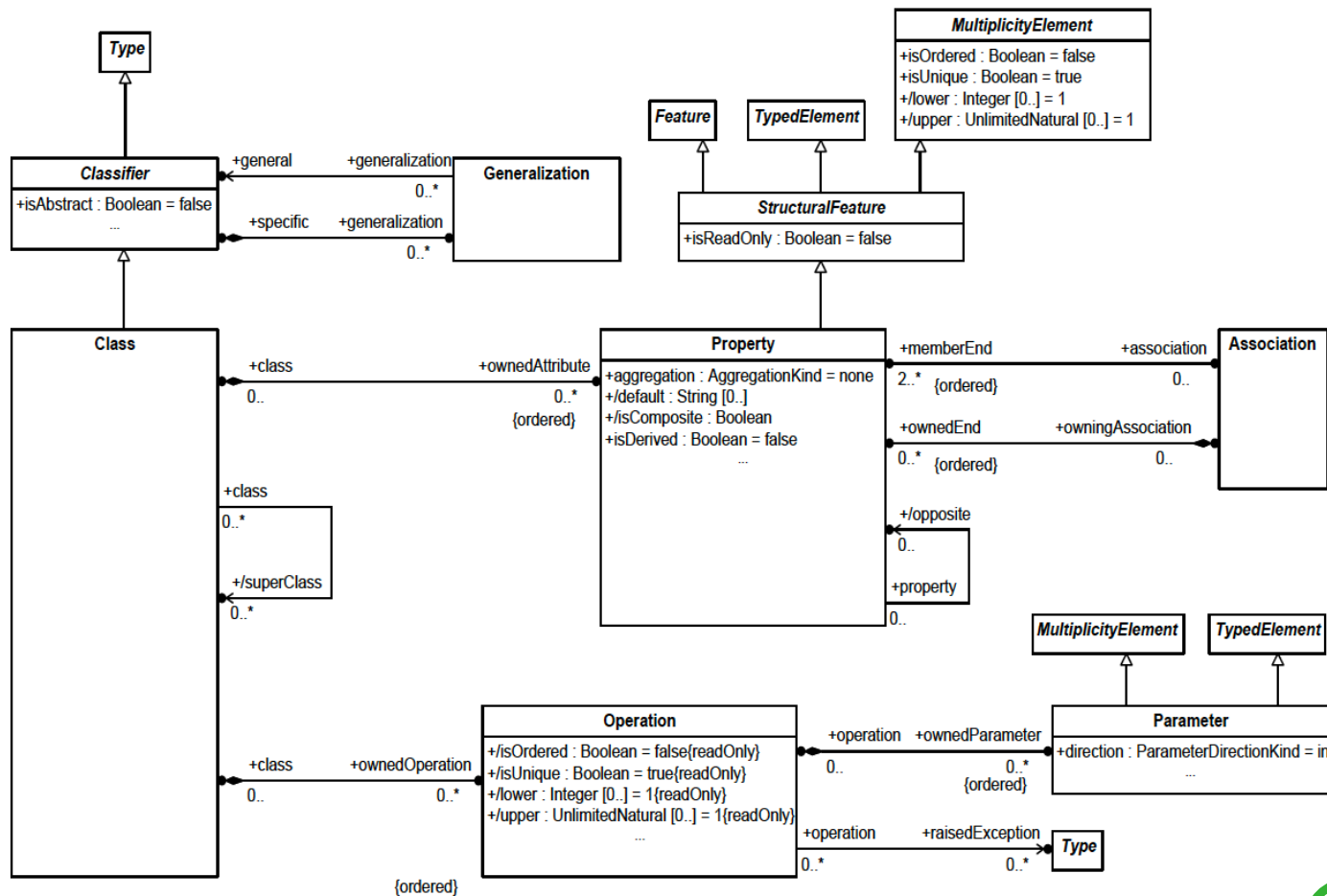
- 
- MOF 2.x consists of **Essential MOF (EMOF)** and **Complete MOF (CMOF)**
 - EMOF corresponds to **facilities found in OOP and XML**
 - Easy to map EMOF models to JMI, XMI, etc.
 - CMOF is used to **specify the metamodels of languages such as, e.g., UML 2**
 - Built from EMOF and the core constructs of UML 2
 - Both EMOF and CMOF are based **on parts of UML 2**

XML-based standard notation for
model serialisation and exchange

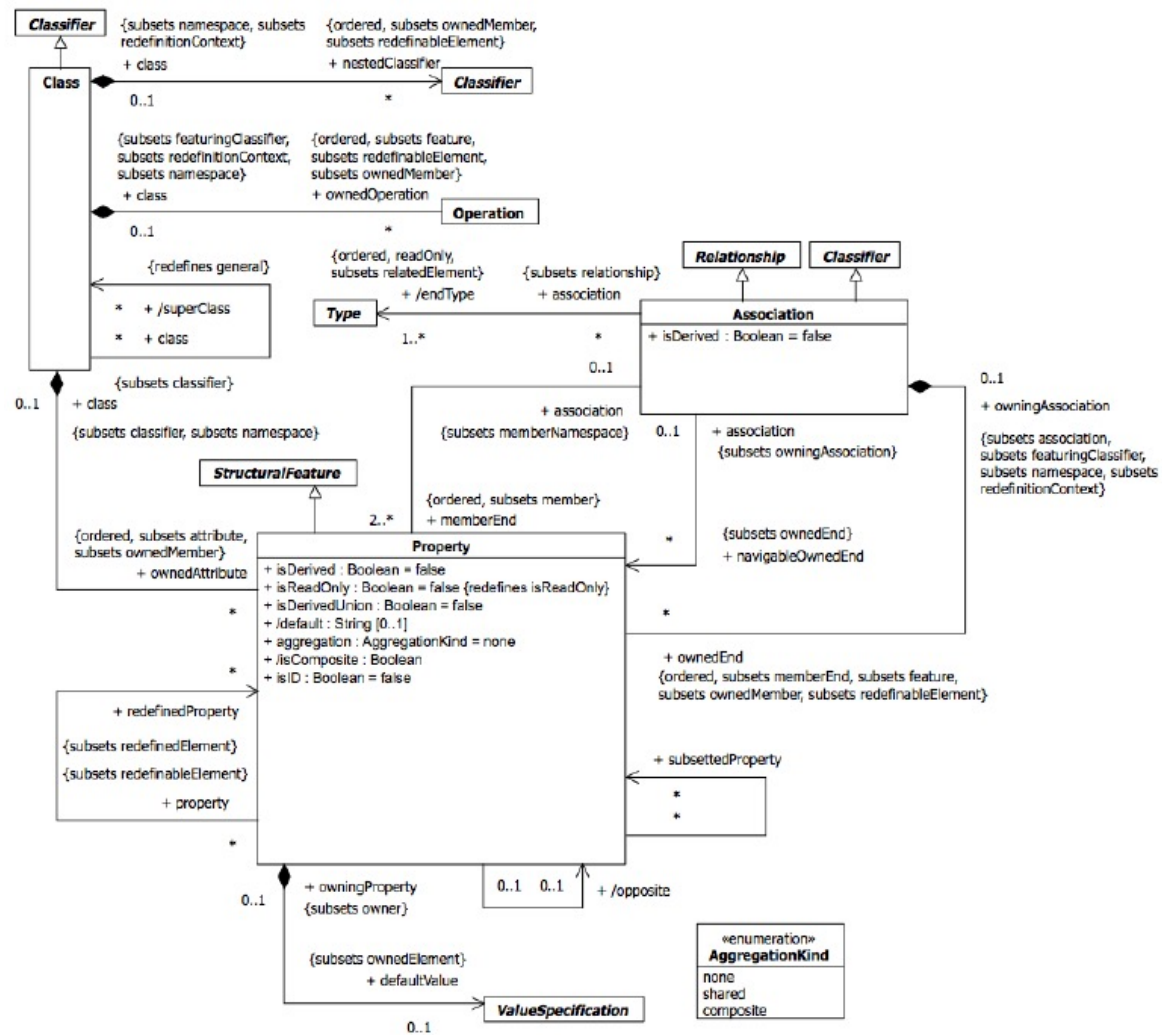
MOF 2.5.1 STRUCTURE



EMOF CORE CLASSES



CMOF CORE CONSTRUCTS



MOF IMPLEMENTATIONS

- 
- Ecore is the most widely known and used, and has been defined in the Eclipse Modelling Framework (EMF)
 - Mostly compatible with MOF 1.x, and allows EMOF metamodels to be imported via XMI

Other environments and tools (historical importance)

- XMF-Mosaic implements ExMOF, which subsets and extends MOF 1.x
- UML2MOF is a transformation from UML metamodels to MOF 1.x metamodels
- Sun MDR implementation
- Commercial implementations from Adaptive, Compuware, MetaMatrix, MEGA, Unicorn



ECLIPSE MODELLING FRAMEWORK (EMF) AND ECORE

- Just about every program **manipulates some data model**
 - Defined in Java, UML, XML Schemas, or some other definition language
- EMF aims to **extract this intrinsic model** from programs and **generate implementation code** from this model
 - Can realise a **tremendous productivity gain**
- Ecore is a metamodel language **similar to the MOF**
→ Ecore and MOF **are not identical** (and **not the same**)!

EMF (ECORE) MODEL DEFINITION

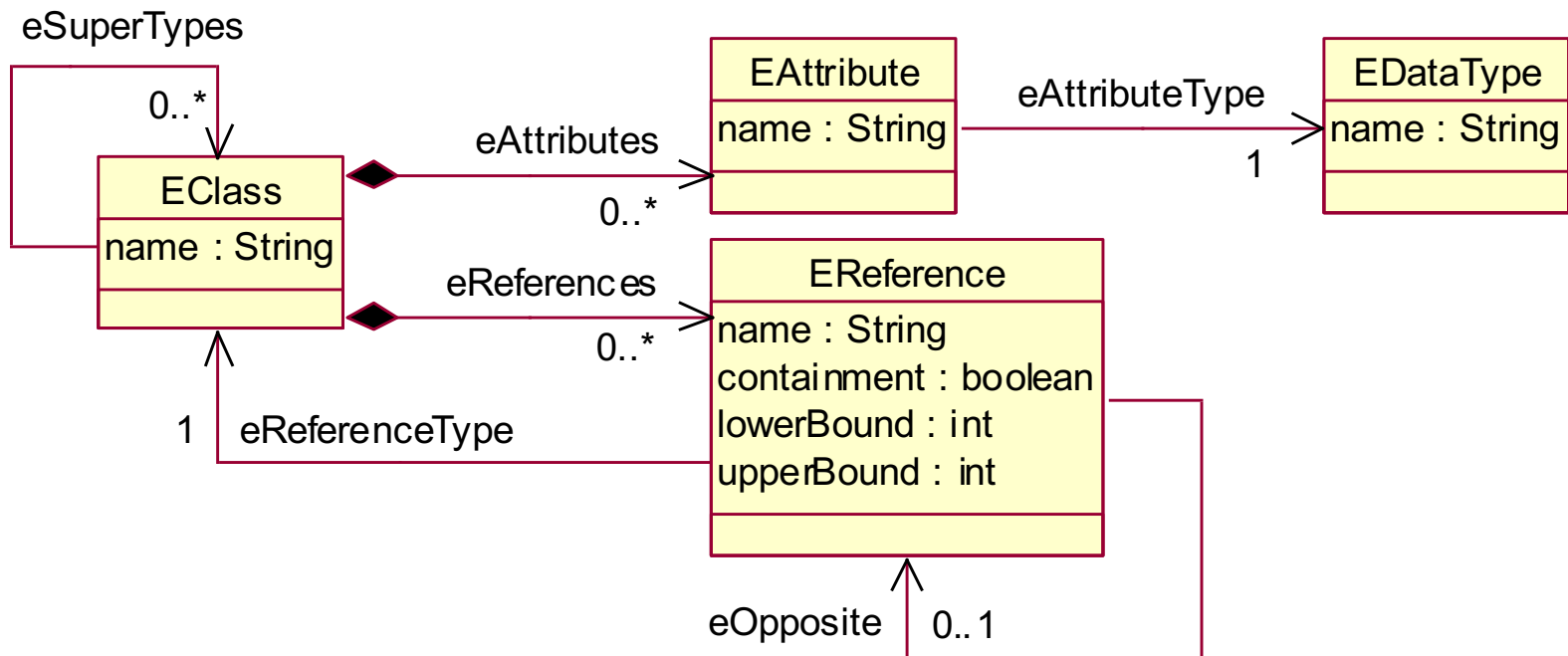
- 
- Specification of application data
 - Attributes of objects
 - Relationships (associations) between objects
 - Operations available on each object
 - Simple constraints (e.g., multiplicity) on objects and relationships
 - Essentially the Class Diagram subset of UML

EMF (ECORE) MODEL DEFINITION

- 
- EMF (Ecore) models can be defined in (at least) three ways
 - Java interfaces
 - UML Class Diagram
 - XML Schema
 - From these artifacts, EMF can generate the others as well as the implementation code
 - In this course, we use the **Ecore tools SDK** to create **Ecore models** by **drawing them** (similarly to UML class diagrams in a UML tool)

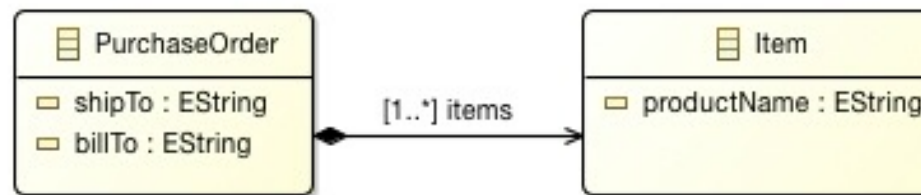
Important!

ECORE MAIN CLASSES

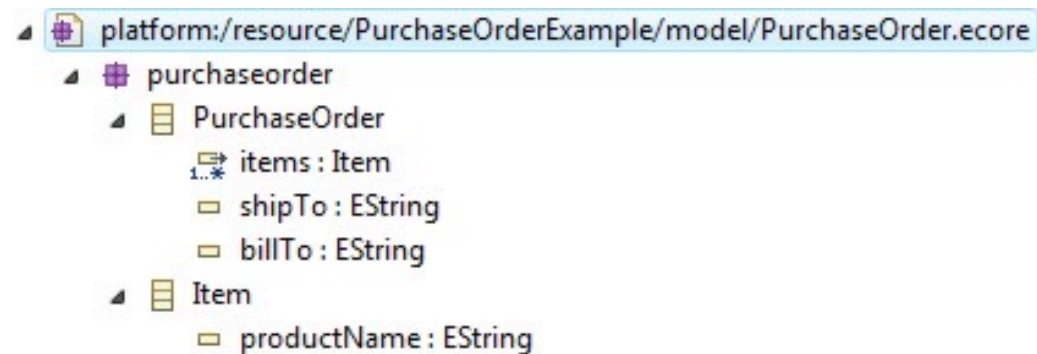


SIMPLE EXAMPLE: PURCHASEORDER

- Ecore diagram drawn with the Ecore editor



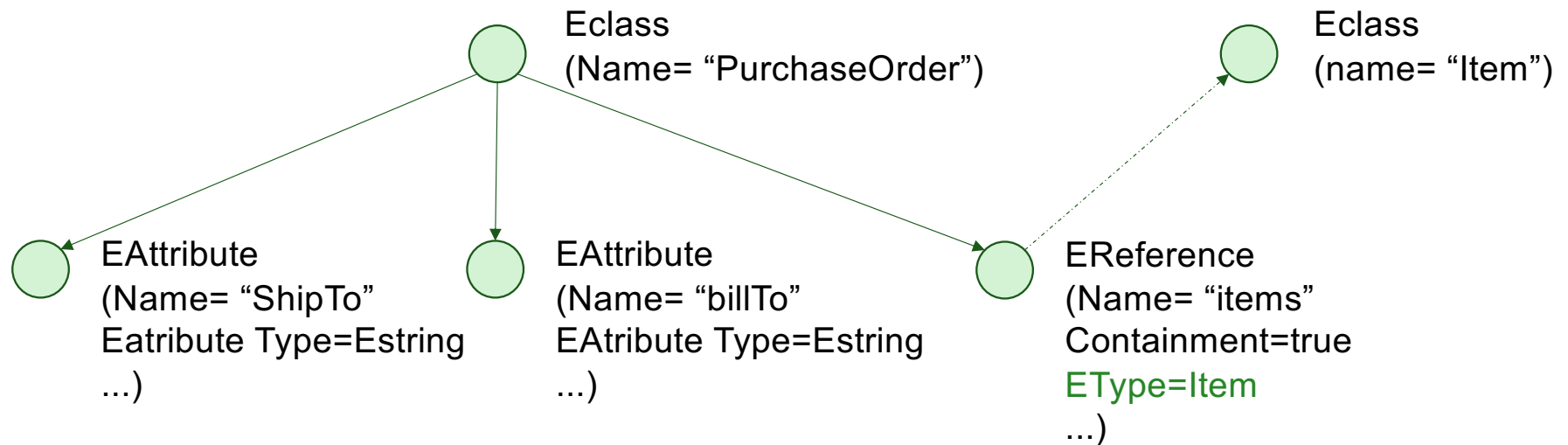
- Ecore model structure



Demo!

PURCHASEORDER Ecore Model

Graph Representation



TAKE-HOME MESSAGES



- In MDE
 - A **metamodel** defines the modelling language used to represent a model
 - A **model** is an **instance** of its metamodel
 - A **metamodel** is an **instance of a metamodel**
→ language to define metamodels
- **MOF** is the **OMG standard metamodel**
- **EMF (Ecore)** is a **de facto standard** used to experiment with models and metamodels

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